

Question

Propose a mechanism for the reaction shown in Figure 1, considering whether specific stereochemistry will form at the hydrogen atom marked with *, and the principle of its formation.

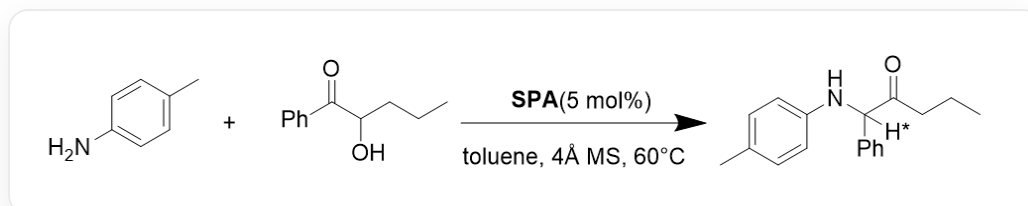


Fig. 1, The reaction in the figure is represented by SMILES as:

CC1=CC=C(N)C=C1.CCCC(O)C(C2=CC=CC=C2)=O>>CCCC(C([H*])
(C3=CC=CC=C3)NC4=CC=C(C=C4)C=O, where the reaction conditions are: SPA (5 mol%), toluene, 4 Å
 MS, 60°C.

The structure of SPA is shown in Figure 2:

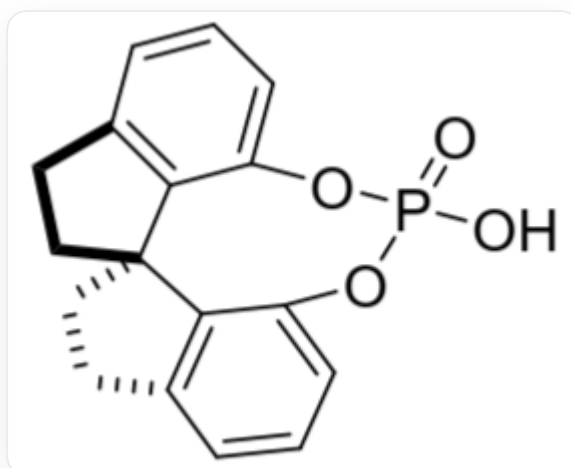


Fig. 2, The molecular structure in the figure is described by SMILES as: O=P1(O)OC2=CC=CC3=C2[C@@](CC3)(CC4)C5=C4C=CC=C5O1

There are the following statements:

1. The reaction of the primary amine with the ketone is later than the elimination reaction

2. The key step of the reaction involves two isomerization reactions
3. In order to further increase the proportion of obtaining a product with specific stereochemistry, all of the oxygen atoms in the phosphate catalyst should be esterified with ligands as much as possible, leaving only one P – OH single bond
4. Stereoselectivity originates from one step in the elimination reaction

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1,2

G. 1,3

H. 1,4

I. 2,3

J. 2,4

K. 3,4

L. 1,2,3

M. 1,2,4

N. 1,3,4

O. 2,3,4

P. 1,2,3,4

Answer

Correct Answer: **C**

Detailed Explanation

Primary amines react with α -hydroxy ketones to form imine compounds, and the imine can undergo tautomerization to form the structure in Figure 3:

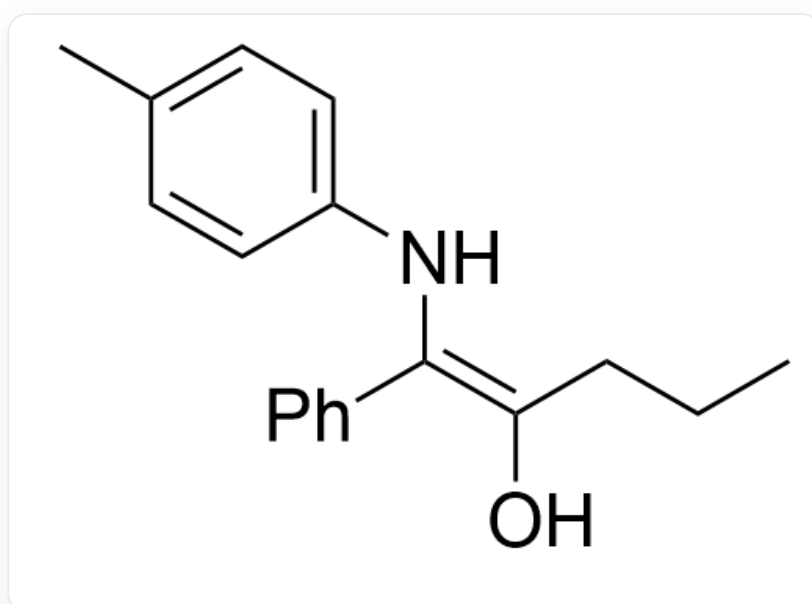


Fig. 3, the molecule in the figure is represented by SMILES as:

CC(C=C1)=CC=C1N/C(C2=CC=CC=C2)=C(CCC)/O

CHECKPOINT

1 PTS

Formation of imine, imine tautomerization to form the following structure described by SMILES:

CC(C=C1)=CC=C1N/C(C2=CC=CC=C2)=C(CCC)/O

This structure is also an enol structure, which can undergo proton transfer isomerization to generate ketone compounds. This process involves the formation of a chiral center, with the proton provided by stereospecific SPA. The $P=O$ of SPA can form a hydrogen bond complex with the hydroxyl group, while $P-OH$ is close to the enol carbon atom, resulting in a one-step concerted proton transfer to generate ketones. In this process, the bulky groups on the phosphate play a stereoselective role, causing protonation to occur only on one side of the double bond plane.

Involving imine and ketone tautomerization, statement 2 is correct.

If the phosphate catalyst does not have $P=O$, it cannot form a hydrogen-bonded catalytic concerted transition state with the carbonyl group, making it difficult to exert stereoselectivity, statement 3 is incorrect.

CHECKPOINT

1 PTS

SPA catalyzes enol proton transfer, $P=O$ can form a hydrogen bond complex with the hydroxyl group, and the bulky groups of the phosphate play a stereoselective role

Stereoselectivity comes from enol protonation, statements 1,4 are incorrect.